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Changes on EMG activation in healthy subjects and incomplete SCI patients following a robot-assisted locomotor training

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***Abstract*—The aim of this study was to understand and measure the lower limbs muscular activation patterns both in healthy and spinal cord injured (SCI) subjects during robot-assisted locomotor exercise. Electromyographic (EMG) activity of four leg's muscles (rectus and biceps femoris, tibialis anterioris and gastrocnemius) was recorded and analyzed at two different percentages of body weight support, three stepping velocities and three different modalities. SCI subjects were recorded also after four weeks training to evaluate the effectiveness of lower limb robot-assisted rehabilitative treatment.**

A multi-factor ANOVA on the integrated muscle activity (IEMG) parameters both in healthy and SCI subjects was performed.

Higher muscular activities both in healthy subjects and SCI patients were found during the exercises using the "DGO active" modality and higher stepping velocities.

A significant increased bilateral muscular activity was observed in each SCI subject after the rehabilitation treatment.

The method proposed to analyze EMG data provides a quantitative description of the lower limb muscular recruitment and can contribute to identify the optimal rehabilitation treatment's conditions.

Keywords- Robotics; Rehabilitation; Spinal cord injury; Motor recovery; Lower limbs.

I. INTRODUCTION

The incidence of spinal cord injury varies in different countries from 10.4 to 83 cases per million population per year [1], [2]. One-third of these patients reported a paralysis of the lower and superior limbs and 50% of them suffer a complete injury.

The mean age of patients who suffered this type of injury is 33 years and the distribution of sex (men/women) is 3.8/1 [3]. In Europe the estimated average annual incidence of spinal cord injury is represented by 20/25 new traumatic cases per one million people. In addition, patients that reported a paralysis of the lower limbs due to diseases (viral, tumor, etc.) must be included, as well. Therefore, the average yearly incidence is represented by 25-30 new cases per one million people [4], [5].

The life expectancy of the survivors has increased, thanks to technical improvements in treatment of acute injuries and complications of neurological dysfunction and therefore the quality of life.

The goal of rehabilitation in patients with neurological diseases is to recover the sensory-motor functions and then to mobilize the lower limbs as soon as possible, repeatedly and in several directions.

During the last two decades, treadmill training has become an established rehabilitation exercise for individuals with locomotor dysfunction such as SCI patients [6]-[8]. Manually assisted walking on a treadmill combined with body weight support (BWS) has been used for more than 10 years in the locomotor rehabilitation of SCI patients and this training improves walking capabilities [8], [9]. Using a robotic driven gait orthosis (DGO), in conjunction with a treadmill and a body-weight support system is possible to obtain potential advantages: (1) patient safety, (2) repeatability of the movements, (3) unlimited duration of training earlier in rehabilitation program and (4) objective functional assessment.

In addition, the integration of robotic systems for lower limbs rehabilitation and virtual reality interfaces, acoustic and visual feedbacks can improve patient's rehabilitative process and increase their motivation [10]. Despite possible limitations on physiological movements, robotic systems for locomotion let patients to execute mass-practiced movements in a highly consistent manner [11].

The objectives of the present study are: (1) to compare the muscular activation patterns both in healthy subjects and SCI patients during robot-assisted locomotor exercise using different velocities, percentages of body weight support and modalities; (2) to identify the most effective strategy for patient-robot cooperation; (3) to measure muscle activities that are needed to conduct a statistic analysis and to do a objective functional assessment; (4) to study the changes in the muscular recruitment after locomotion rehabilitation using a robotic device and evaluate the effectiveness of the rehabilitative treatment using robotic systems.

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II. METHODS

A. Participants

Five control subjects without neurological deficit (mean age 23.6±1.3, one man and four women) and five incomplete chronic SCI subjects (mean age 42.4±10.6, three men and two women) were recruited. The SCI patients were evaluated (Table I) using the ASIA scale, which measures the motor function (ranging from A to maximum value E) [12], [13] and the SCIM (maximum value 100) [14].

TABLE I. CHARACTERISTICS OF SCI SUBJECTS

ID	*Sex*	*Age*	*ASIA*		*SCIM*
P1	M	33	L1	D	64
P2	M	44	T4	C	65
P3	F	58	T10	D	70
P4	F	32	C6	D	65
P5	M	45	T8	D	84

B. Test Protocol

The Lokomat system (Hocoma AG, Volketswil, Switzerland) [4], [15] used in this study, is a bilateral exoskeleton device with two degrees of freedom per leg for knee and hip rotation (Figure 1).

The Lokomat training is carried out through a position controller that tracks reference hip and knee joint-angle trajectories in the sagittal plane [1]. Hip and knee joints are actuated by DC motors and linear ball screw, which are built into the exoskeletal structure. In this structure, position and force sensors are integrated to provide input for the controller. The entire system is mounted on a treadmill that follows the step in the stance phase of the stride cycle. The subject's weight is unloaded by a mechanical system that can vary from 5 to 80 kg. The Lokomat system is characterized by a parallelogram structure aimed at stabilizing the subject's trunk. Two safety press buttons let the patient to stop the DGO motion. The system is provided with adjustable components in order to take into account the anthropometric characteristics of subjects.

Each subject, after providing informed consent, is asked to perform the exercises planned by the clinical protocol, composed by 18 trials, each lasting 60 seconds. Each trial is characterized by BWS percentage (30% or 60%), treadmill speed (1 km/h, 1.6 km/h and 2.4 km/h) and modality of human-robot interaction ("DGO active", "DGO passive" or "without DGO"). During the "DGO active" modality, the subject is asked to actively walk as normal as possible within the DGO in order to cooperate with the device; during the "DGO passive" modality, stepping movements are induced by the Lokomat in order to reduce the physical interaction between the patient and the robotic system; during the "without DGO" modality the subject is asked to walk on the treadmill without the robotic DGO.

During the training, two customized foot-switches (Figure 2) are taped directly to the subject's heels to broken up the stride cycle into two distinct phases: stance and swing. Each stride was considered as the period between two successive hell-strikes on the same leg.

Figure 1. A subject during the locomotor training using the Lokomat system.

In each trial, surface EMG electrodes were attached to both subject's lower limbs on rectus femoris (RF), biceps femoris (BF), tibialis anterioris (TA) and gastrocnemius (GA). The EMG activities were recorded and both the EMG data and foot-switch signals were sent to two 8 channels transceivers (TeleMyo 2400, Noraxon USA, Inc., Scottsdale, AZ, USA) that are attached to a belt fixed to the subject's waist. The data are transmitted to a laptop for data storing.

Figure 2. Customized foot-switch with the relative lodge.

C. Robot-Assisted Therapy

The SCI subjects underwent a Lokomat training for four weeks, five sessions per week. In each session they received 60 minutes of robot-assisted therapy, using a decreasing percentage of BWS ranging from 50% to 15% and speeds from 1.5 km/h to 2.0 km/h.

The subjects were assessed before ("Pre-training") and after the four weeks training ("Post-training"), through both a clinical assessment and the same test protocol applied to healthy subjects. A clinical evaluation was performed by an experienced physiatrist. Gait speed was assessed through 10-meter walk test (10 MWT), 6-minute walk test (6 MWT) and Timed Up & Go (TUG) test [16] before and after the robot-assisted training.

D. Data Analysis

The EMG signals were acquired with a sampling rate of 3000 Hz using an 8th order Butterworth low-pass anti-alias filter set at 1000 Hz, using two customized foot-switches. EMG data were processed using custom routines implemented under the Matlab environment (Matworks Inc., Natick, MA, USA). For each subject's recording, the following steps were applied:

1) Heel-strike detection. Foot-switches data were extracted, filtered and rectified. Data from the foot-switches were used to divide the EMG signals into a number of slices equal to the number of steps performed during each specific locomotion exercise;

2) Signal processing. For each muscle, exercise and subject, the EMG signals (μV) were extracted and divided using the foot-switches data;

3) Signal filtering. The EMG signals were high-pass filtered at 20 Hz, rectified and low-pass filtered at 5 Hz using a 4th order Butterworth filter [17];

4) Signal averaging and normalizing. For each exercise, the EMG signals were averaged, normalized and then expressed as a percentage of the total gait cycle (e.g., 0-100%);

5) Data structures. For each muscle, exercise and subject, the averaged normalized EMG signals were stored into a multidimensional data structure;

6) Statistical analysis. From EMG signals, the integrated value of EMG signals (IEMG) was computed to quantify the muscular recruitments. A statistic analysis based on the IEMG parameters was performed using a multi-factor ANOVA, both in healthy subjects and SCI patients. The following formulas were used for the computation of the IEMG during the stance phase ($IEMG_1$) and swing phase ($IEMG_2$), where t_1 stands for the 0%, t_2 the 60%, t_3 the 61% and t_4 the 100% of stride cycle.

$$IEMG_1 = \int_{t_1}^{t_2} SGN_{EMG}\,dt$$

$$IEMG_2 = \int_{t_3}^{t_4} SGN_{EMG}\,dt \qquad (1)$$

On the group of healthy subjects, three statistical tests were performed using a one-factor analysis of variance (ANOVA), where the fixed factor was (1) the speed of ambulation, (2) the modality and (3) the BWS percentage, whereas the dependent variable was the integrated normalized EMG activity for each respective phase and muscle.

On the group of SCI subjects, three statistical tests were performed using a two-factors ANOVA, where the fixed factors were the phase of treatment and, alternatively, (1) the speed of ambulation, (2) the modality and (3) the BWS percentage, whereas the dependent variable was the integrated normalized EMG activity for each respective phase and muscle.

III. Results

A. Healthy Subjects

Figure 3 shows the mean EMG activities on the right leg (similarly observed on the left leg), using 30% BWS and 60% BWS, in healthy subjects. In both cases, the "DGO passive" modality shows a low recruitment of proximal muscles (RF and BF) and a high activity for the distal muscles (GA and TA). In the "DGO active" modality, a clear increased activation of all the muscles is observed. During the "without DGO" modality, the recruitment of RF, BF and GA is similar to that observed during the "DGO passive" modality; on the other hand, the recruitment of TA is similar to that observed during the "DGO active" modality.

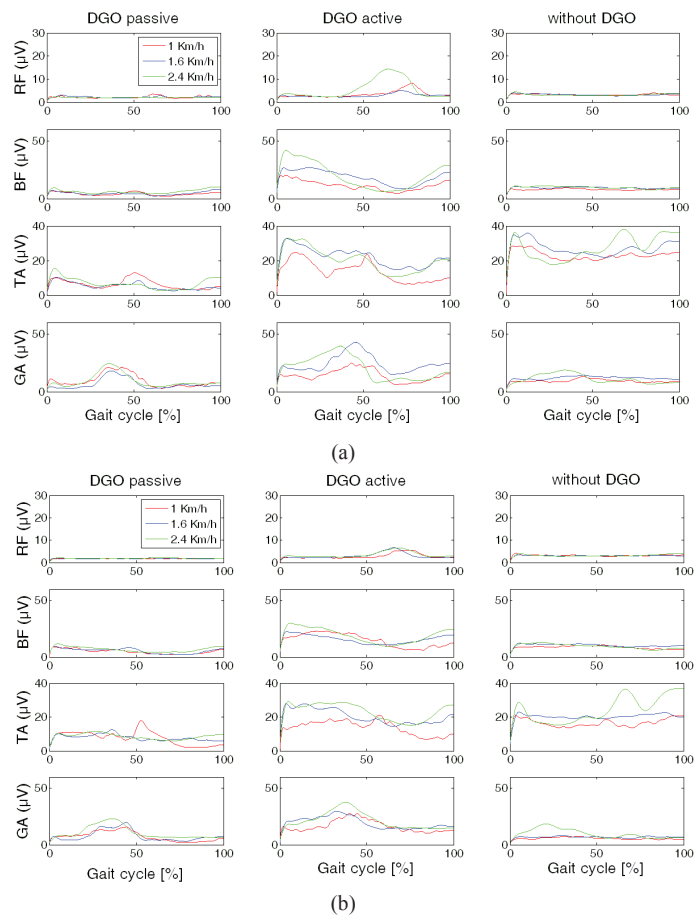

Figure 3. Healthy subject's averaged EMG activity of four muscles of the right lower limb (RF; BF; TA; GM) during the 30% BWS (a) and 60% BWS (b), across three different speeds for both robot-assisted ("DGO passive" and "DGO active") and treadmill exercise ("without DGO").

The statistical analysis using the treadmill speed as fixed factor (Figure 4a) does not show statistically significant results, but a trend to increasing muscular activity is observed at higher speeds.

The statistical analysis using the modality as fixed factor (Figure 4b) shows several statistically significant results:

higher recruitments are observed during the "DGO active" modality in BF and GA; only in TA during the "without DGO" modality. No statistically significant results were observed using the BWS percentage as fixed factor: the recruitment in the two cases (30% and 60% BWS) are quite similar.

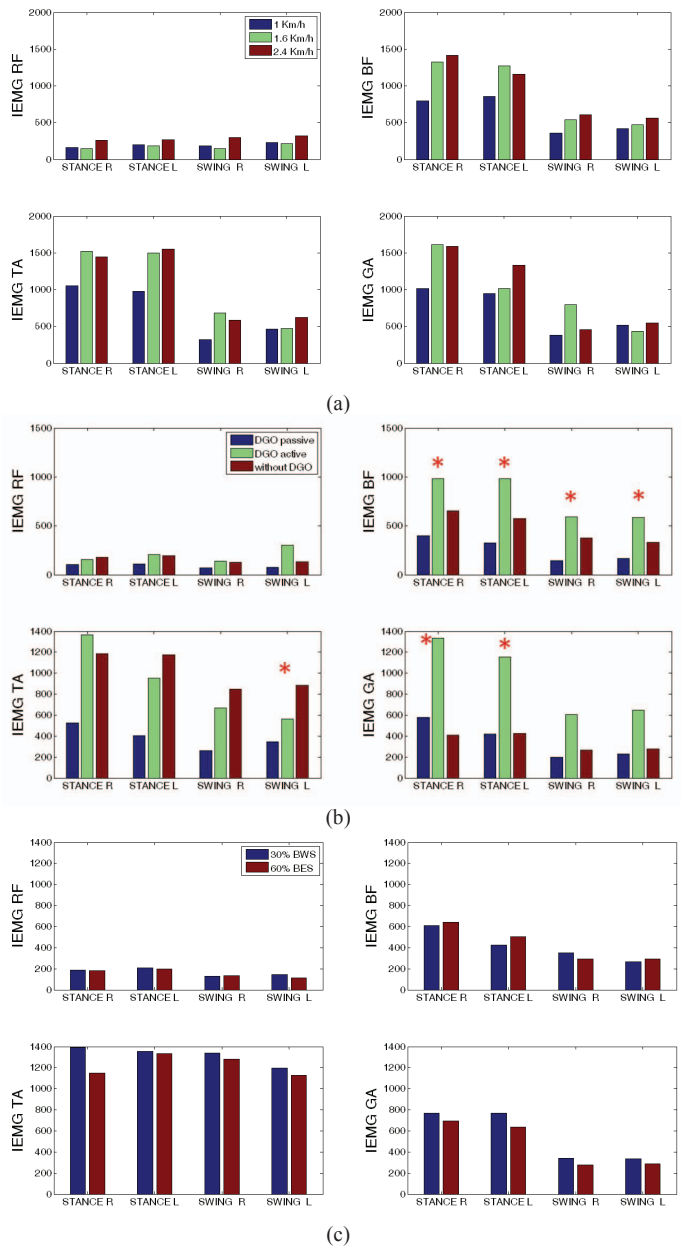

Figure 4. Mean integrated muscle activity from the group of healthy subjects for each of the four muscles, during the two gait phases (stance and swing) on the right (R) and left side (L): (a) 30% BWS, "DGO active" and different treadmill speeds; (b) 60% BWS, 1.6 km/h speed and different modalities; (c) 2.4 km/h speed, "without DGO" and different BWS (* $p<0.05$).

B. *Lokomat Therapy*

An improvement in the ambulatory function in four SCI subjects was observed through the clinical assessment. In fact, as shown in Table II, after the robot-assisted training, the distance in 6 MWT increased and the time in TUG decreased. In subject P1 an improvement in the 10 MWT speed can be observed, as well.

TABLE II. SCI SUBJECTS' ASSESSMENT MEASURES

	ID				
	P1	***P2***	***P3***	***P4***	***P5***
10 MWT PRE (km/h)	0.72	0.71	0.15	1.15	N/A
10 MWT POST (km/h)	1.89	0.95	0.43	1.17	2.9
Δ MWT (km/h)	**1.17**	**0.24**	**0.28**	**0.02**	**N/A**
6 MWT PRE (m)	76.2	27.4	30.75	46.84	N/A
6 MWT POST (m)	167.7	43	47	106.2	249.6
Δ MWT (m)	**91.5**	**15.6**	**16.25**	**59.36**	**N/A**
TUG PRE (s)	42	58	78	32.72	N/A
TUG POST (s)	27	49.37	65	26.7	15
Δ TUG (s)	**-15**	**-8.63**	**-13**	**-6.02**	**N/A**

The EMG signals (Figure 5) and values of the IEMG parameters (Figure 6) show an improvement in the muscular recruitment in three different trials on subject P1: (1) 30% BWS, "DGO active", 1 km/h (Figure 5a), (2) 30% BWS, "DGO active", 1.6 km/h (Figure 5b) and (3) 30% BWS, "without DGO", 1 km/h (Figure 5c).

The plots of EMG signals in three representative trials reported for subject P1 show a poor activation in all muscles before the robot-assisted therapy (Figure 5, left column). The EMG recording during the three exercises after the robot-assisted therapy shows a good bilateral muscular recruitment (Figure 5, right column).

The values of the IEMG parameters in the three different trials on subject P1 at the post-treatment are higher than those observed at the pre-treatment (Figure 6).

The statistical analysis performed using the treadmill speed and the phase of treatment as fixed factors (Figure 7a) shows statistically significant results: an increased muscular activity is observed after the four weeks of therapy. The statistical analysis performed using the modality and the phase of treatment as fixed factors (Figure 7b) shows statistically significant results.

The EMG recruitment in the SCI subject observed at the post-treatment (Figure 5, right column) tends to be similar to that observed in healthy subjects (Figure 3).

IV. DISCUSSION

In our study, the group of healthy subjects was analyzed in order to collect EMG reference signals according to the conditions described by the clinical protocol. In a previous study, the EMG signals during the robot-assisted exercise were recorded without any BWS [6]: in our study, 30% and 60% BWS were used on healthy subjects, with the aim of comparing their EMG activity to that of neurologically impaired subjects, who need to be weight supported.

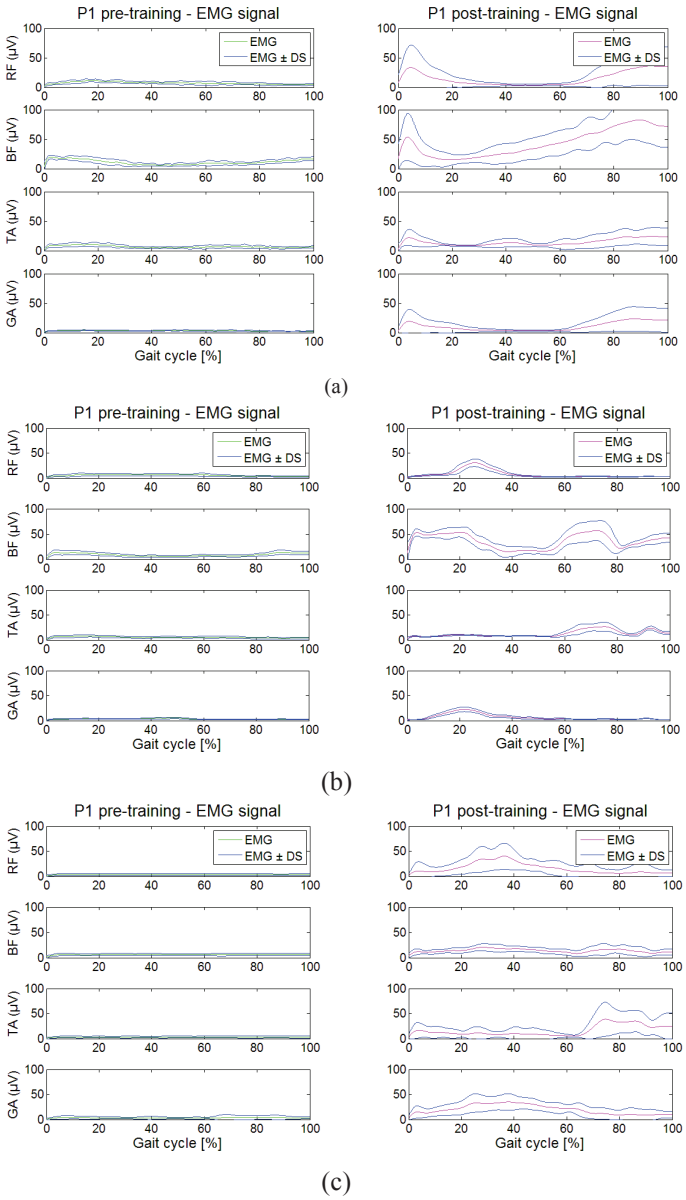

Figure 5. EMG activity of four muscles of the right leg (subject P1): (a) 30% BWS, "DGO active", 1 Km/h speed; (b) 30% BWS, "DGO active", 1.6 Km/h speed; (c) 30% BWS, "without DGO", 1 Km/h speed.

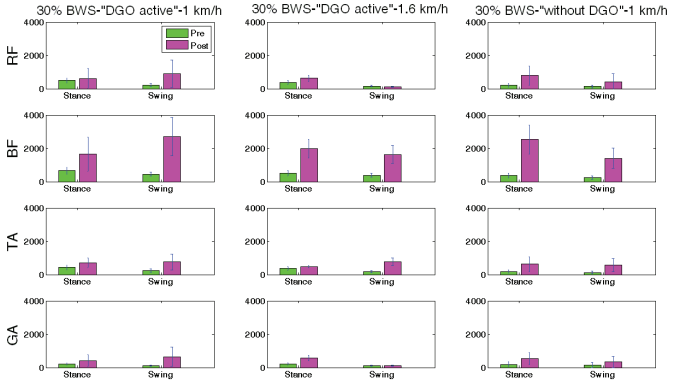

Figure 6. IEMG values of four muscles (RF, BF, TA, GA) and standard deviation on the right leg during three different trials (subject P1): pre-treatment (green); post-treatment (purple).

The muscular activations found on healthy subjects both during the "DGO active" and "DGO passive" modality (60% BWS) show a trend which is similar to that presented in a previous study [18]: in fact, the muscular activations are found in the same ranges of the gait cycle.

We did not found any statistically significant correlations between changes in IEMG parameters on healthy subjects in the case of treadmill speed as independent variable: such finding could be due to relatively low values of the treadmill speed for healthy subjects.

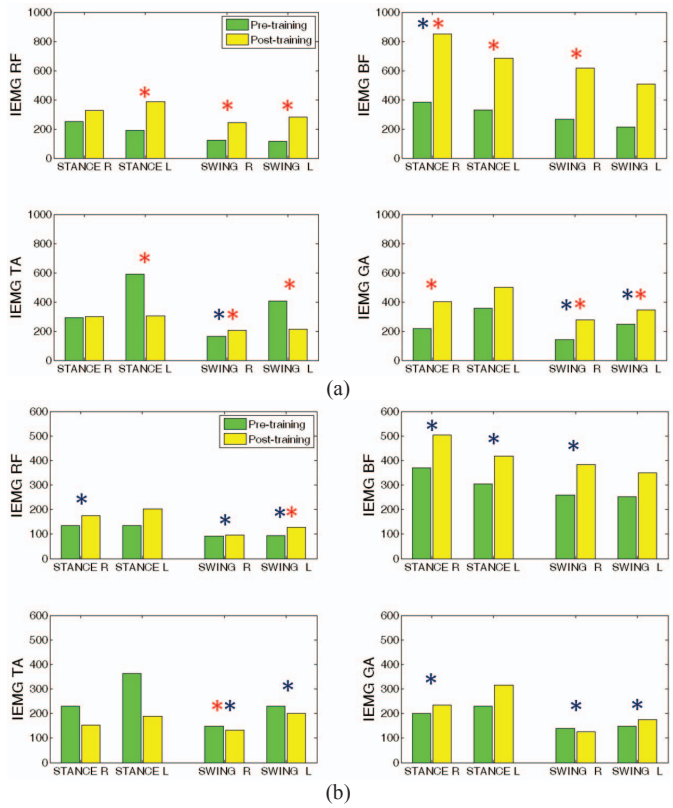

Figure 7. Mean integrated muscle activity from the group of SCI subjects for each of the four muscles during pre-training (green) and post-training (yellow) during the two gait phases (stance and swing) on the right (R) and left side (L): (a) 60% BWS, "DGO active" and different treadmill speeds; (b) 30% BWS, 1.6 km/h speed and different modalities. Fixed factors: red, phase of treatment; blue, (a) treadmill speed, (b) modality.

The activations in muscles observed in subject P1 after the four weeks of therapy in the "DGO active" modality, 30% BWS and speed of 1.6 Km/h (Figure 5b, right column) are found in the same ranges of the gait cycle as in a previous study [9]:

- RF activity from 20% of the gait cycle to 40%.
- BF activity from 0% of the gait cycle to 30% and from 60% to 80%.
- TA activity from 70% of the gait cycle to 90%.
- GA activity from 10% of the gait cycle to 30%.

The statistical analysis performed on the group of SCI subjects highlights which factors have a significant influence on the change of the IEMG values after the treatment: the

treadmill speed and modality of interaction. In the case of BWS as independent variable, no statistically significant results in the IEMG parameters were found, both in healthy subjects and SCI patients. This finding is different from previous results obtained using BWS on the treadmill, without robotic system [19].

At the end of the four weeks therapy, in addition to the "DGO active" modality, even in the "without DGO" modality (Figure 5c and Figure 6, right column), an increased muscular recruitment was observed in each SCI subject, thus supporting the evidence that the robot-assisted training contribute to recover some muscular recruitment, which was absent at the beginning of the therapy.

A spontaneous increase in muscle recruitment is unlikely as SCI subjects recruited in this study were functionally stable from more than one year.

Anyway, a clear evidence whether the effect of the increase of EMG activity can be really traced back to the specific training is still to be clarified through the comparison with a group of patients using an ABA scheme (i.e., active functional training-conventional therapy-active functional training).

V. Conclusion

After the rehabilitation robot-assisted treatment, the SCI patients analyzed in this study show an increased bilateral muscular activity. Improvements in their locomotion function were observed, as well. These results provide evidence in support of the benefic patient-robot interaction.

The robotic device, used in conjunction with a system for the acquisition of EMG data, has proved to be a valuable tool for objectively quantify the muscle recruitment and highlight the effects of the proposed treatment.

For each subject, the proposed analysis lets to identify (1) which muscles require additional rehabilitation treatment and (2) the optimal conditions of the training. Such approach may lead to a personalization of the treatment, adjusting the percentage of BWS, treadmill speed and modality of cooperation, according to the changes occurring during the motor recovery of each subject.

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